Seokjun Ryu

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Research Interests

Offensive security researcher and CTF player focused on **AI Security**, **Windows Kernel Exploitation**, and **Reverse Engineering**, with vulnerability discovery across **AI/ML frameworks and AI agent platforms** (vLLM, OpenClaw, Flowise) and **LLM-based multi-agent systems** for automated vulnerability research.

Education

Soongsil University — **Seoul, South Korea**
B.S. in Electronic Engineering (*IT Convergence*) — *Mar. 2022 – Present*

KITRI WhiteHat School
3rd Cohort, Vulnerability Analysis Track (*Completed*) — *Mar. 2025 – Sep. 2025*
- Project: Windows kernel driver 1-day analysis and exploitation (CVE-2024-35250)

POC Security
Hacking Camp 29th (*Completed*) — *Aug. 2024*

Work Experience

AI Safety Center, Soongsil University — **Seoul, South Korea**
Undergraduate Research Intern, AI for Security — *Jan. 2026 – Present*

Publications

Domestic Conference

Analysis of Vulnerability Attack Pathways in AI/ML Frameworks Based on Public CVEs and a Lightweight LLM Prompting-Based Bug Hunting Methodology — May 2026
Korea Institute of Information Security & Cryptology Summer Conference 2026 (***Best Paper Award***)
Seokjun Ryu, Sohee Park, Domin Ok, and Daeseon Choi

Talks

AI for Security for AI: Hacking AI Agents with AI Agents — Aug. 2026
Hacking Camp 33rd, POC Security
Seokjun Ryu and Taeyoung Jeong

Vulnerability Disclosures

CVE-2026-100557 OpenClaw — *Disclosed via GHSA*
CVE-2026-62203 OpenClaw — *Disclosed via GHSA*
CVE-2026-35626 OpenClaw — *Disclosed via GHSA*
CVE-2026-34755 vLLM — *Disclosed via Huntr*
GHSA-c2p8-**-**** OpenClaw — *Disclosed via GHSA, CVE pending*
GHSA-rwm5-**-**** OpenClaw — *Disclosed via GHSA, CVE pending*
GHSA-rj8f-**-**** OpenClaw — *Disclosed via GHSA, CVE pending*
GHSA-5968-**-**** Flowise — *Disclosed via GHSA, CVE pending*

Projects

Development of Multi-Agent Vulnerability Research System May 2026 – Aug. 2026

- Achieved **91.0% pass@1** on CyberGym Level 1 (1,372/1,507 tasks) by building **mneme**, a multi-agent system with structured causal memory and execution-feedback refinement, matching Microsoft MDASH and beating the 66.6% same-model (Claude Opus 4.6) baseline by 24.4 points. [Report] [CyberGym] [Press]
- Directed the Mythos+ Task Force as **TF Lead**, owning operations, agent architecture design, and the evaluation framework.

Development of Robust AI and Distributed Attack Detection Technologies for Edge AI Security Mar. 2026 – Present

- Built the **frontend and backend** of an IITP-funded Edge AI Vulnerability Analysis System prototype, a unified dashboard spanning adversarial-example, adversarial-patch, and voice attacks across image-classification, object-detection, and speech-recognition models.
- Automated model/attack selection, simulated attack execution, on-demand attack-data generation, and result reporting.

Automated Agent-Targeted Bug Hunting with AI Agents (Agent-Zero) Mar. 2026 – Aug. 2026

- Discovered **38 valid vulnerabilities** across OpenClaw, n8n, and qbee-agent by building **CrayFisher**, an agentic zero-day bug-hunting agent based on an OpenClaw 1-day vulnerability taxonomy targeting AI-agent codebases.
- Led the project as **Team Lead** (ASC hacking club).

Windows Kernel Privilege-Escalation 1-Day Analysis and Exploitation (CVE-2024-35250) Mar. 2025 – Sep. 2025

- Developed a working local privilege-escalation exploit for **CVE-2024-35250**, gaining a SYSTEM shell by building a kernel read/write primitive via `PreviousMode` overwrite under kASLR/SMEP/kCFG and overwriting the process token.
- Identified the root cause with IDA and WinDbg: a kernel streaming driver re-issues an IOCTL with the new IRP's `RequestorMode` hardcoded to `KernelMode`, invoking a user-supplied function pointer without validation (trust-boundary violation).
- Led **root-cause analysis and technical documentation** at WhiteHat School (KITRI). *Stack: C++, Assembly.*

Honors and Awards

Awards

Minister of Science and ICT Award Nov. 2026 (Expected)

CVD/VDP Pilot Program (*Security Vulnerability Reporting, Remediation & Disclosure*), *Korea Internet & Security Agency* (*KISA*)

Best Paper Award (National Security Research Institute Director's Award) May 2026

Korea Institute of Information Security & Cryptology Summer Conference 2026

CTF Competitions

Teams: **ASC** (ssu-asc.com) • **Fermion** (project-plasma.org) • **Physical Lab** (phylab.space)

HITCON CTF 2026 *Team Korea Hacking Club Alliance* **5th**

Cyber Conflict Exercise 2026 *Team 클로드짱의 성능은 망가져있다* **Finalist (6th)**

Hacksium Busan 2026 *Team 나는야임영웅* **Finalist**

Hacktheon International University Student Cyber Security Competition 2026 *Team 나는야임영웅* **Finalist**

HyperSonic CTF *Team ASC* **2nd**

BITSCTF 2026 *Team Fermion* **4th**

HSpace Club League 25 *Team ASC* **5th**